

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12403261
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Hourmand; Yannick et al.

Auto-injector

Abstract

According to the invention, an auto-injector for administering a dose of a liquid medicament (M) comprises of a substantially cylindrical housing arranged to contain a pre-filled syringe filled with the medicament (M), a needle shroud slidably arranged with respect to the housing and adapted to rest on the skin of a patient receiving an injection, a releasable drive means arranged within the housing that is capable of, upon release, translating the needle shroud in a proximal direction (P) towards a safe position (PS) and a rotating collar rotatably arranged within the housing. The needle shroud in the safe position (PS) surrounds the injection needle after the injection has been carried out. The rotating collar engages the needle shroud in a manner that forces the rotating collar to rotate within the housing when the needle shroud is translated in the proximal direction (P).

Inventors: Hourmand; Yannick (Haslingfield, GB), Barrow-Williams; Timothy Donald (Herts, GB), Ekman; Matthew (Cheshire, GB)

Applicant: Sanofi-Aventis Deutschland GMBH (Frankfurt am Main, DE)

Family ID: 43837882

Assignee: Sanofi-Aventis Deutschland GMBH (Frankfurt am Main, DE)

Appl. No.: 19/025123

Filed: January 16, 2025

Prior Publication Data

Document Identifier	Publication Date
US 20250152827 A1	May. 15, 2025

Foreign Application Priority Data

EP	10196067	Dec. 21, 2010
----	----------	---------------

Related U.S. Application Data

continuation parent-doc US 18498416 20231031 PENDING child-doc US 19025123
continuation parent-doc US 16909107 20200623 US 11833331 20231205 child-doc US 18498416
continuation parent-doc US 15669571 20170804 US 10729853 20200804 child-doc US 16909107
continuation parent-doc US 13993528 US 9724472 20170808 WO PCT/EP2011/073502 20111221
child-doc US 15669571
us-provisional-application US 61432255 20110113

Publication Classification

Int. Cl.: A61M5/20 (20060101); A61M5/50 (20060101); A61M5/32 (20060101)

U.S. Cl.:

CPC A61M5/2033 (20130101); A61M5/50 (20130101); A61M2005/206 (20130101);
A61M2005/208 (20130101); A61M5/326 (20130101)

Field of Classification Search

CPC: A61M (25/0169); A61M (25/0172); A61M (2025/0177); A61M (2025/018); A61M
(2025/0183); A61M (2025/0188); A61M (25/01); A61M (25/09); A61M (2025/1045)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3712301	12/1972	Sarnoff	N/A	N/A
4964866	12/1989	Szwarc	N/A	N/A
5478316	12/1994	Bitdinger et al.	N/A	N/A
5480387	12/1995	Gabriel et al.	N/A	N/A
5559309	12/1995	Zabler et al.	N/A	N/A
5599309	12/1996	Marshall et al.	N/A	N/A
6004297	12/1998	Steenfeldt-Jensen et al.	N/A	N/A
6277099	12/2000	Strowe et al.	N/A	N/A
6371939	12/2001	Bergens et al.	N/A	N/A
6454743	12/2001	Weber	N/A	N/A
6620137	12/2002	Kirchhofer et al.	N/A	N/A
6743205	12/2003	Nolan, Jr. et al.	N/A	N/A
7195616	12/2006	Diller et al.	N/A	N/A
7291132	12/2006	DeRuntz et al.	N/A	N/A
7297135	12/2006	Jeffrey	N/A	N/A
7341575	12/2007	Rice et al.	N/A	N/A
7597685	12/2008	Olson	N/A	N/A
7678085	12/2009	Graf	N/A	N/A
7717877	12/2009	Lavi et al.	N/A	N/A
7771398	12/2009	Knight et al.	N/A	N/A

7918824	12/2010	Bishop et al.	N/A	N/A
7976494	12/2010	Kohlbrenner et al.	N/A	N/A
8038649	12/2010	Kronestedt	N/A	N/A
8048037	12/2010	Kohlbrenner et al.	N/A	N/A
8062255	12/2010	Brunnberg et al.	N/A	N/A
8083711	12/2010	Enggaard	N/A	N/A
8313465	12/2011	Harrison	N/A	N/A
8323238	12/2011	Cronenberg et al.	N/A	N/A
8357125	12/2012	Grunhut et al.	N/A	N/A
8361025	12/2012	Lawlis et al.	N/A	N/A
8366680	12/2012	Raab	N/A	N/A
8376993	12/2012	Cox et al.	N/A	N/A
8376997	12/2012	Hogdahl et al.	N/A	N/A
8403883	12/2012	Fayyaz et al.	N/A	N/A
8409138	12/2012	James et al.	N/A	N/A
8409141	12/2012	Johansen et al.	N/A	N/A
8409148	12/2012	Fiechter et al.	N/A	N/A
8439864	12/2012	Galbraith et al.	N/A	N/A
8491538	12/2012	Kohlbrenner et al.	N/A	N/A
8568359	12/2012	Carrel et al.	N/A	N/A
8617109	12/2012	Kronestedt et al.	N/A	N/A
8617124	12/2012	Wieselblad et al.	N/A	N/A
8632507	12/2013	Bartha	N/A	N/A
8647299	12/2013	Stamp	N/A	N/A
8684969	12/2013	Moller et al.	N/A	N/A
8708973	12/2013	Holmqvist	N/A	N/A
8734394	12/2013	Admas et al.	N/A	N/A
8734402	12/2013	Sharp et al.	N/A	N/A
8758292	12/2013	Tschirren et al.	N/A	N/A
8808250	12/2013	Ekman et al.	N/A	N/A
8808251	12/2013	Raab et al.	N/A	N/A
8821451	12/2013	Daniel	N/A	N/A
8834431	12/2013	Kohlbrenner et al.	N/A	N/A
8840591	12/2013	Raab et al.	N/A	N/A
8882723	12/2013	Smith et al.	N/A	N/A
8911411	12/2013	Nielsen	N/A	N/A
8939934	12/2014	Brereton et al.	N/A	N/A
8945063	12/2014	Wotton et al.	N/A	N/A
8956331	12/2014	Johansen et al.	N/A	N/A
8961473	12/2014	Heald	N/A	N/A
8968256	12/2014	Raab	N/A	N/A
8968258	12/2014	Nzike et al.	N/A	N/A
8992484	12/2014	Radmer et al.	N/A	N/A
8992487	12/2014	Eich et al.	N/A	N/A
9005160	12/2014	Karlsson et al.	N/A	N/A
9011386	12/2014	Kronestedt et al.	N/A	N/A
9011387	12/2014	Ekman et al.	N/A	N/A
9022991	12/2014	Moeller	N/A	N/A
9022994	12/2014	Moser et al.	N/A	N/A
9044548	12/2014	Miller et al.	N/A	N/A

9044553	12/2014	James et al.	N/A	N/A
9057369	12/2014	Kohlbreuner et al.	N/A	N/A
9061104	12/2014	Daniel	N/A	N/A
9067024	12/2014	Roberts et al.	N/A	N/A
9089652	12/2014	Nzike et al.	N/A	N/A
9108002	12/2014	Markussen	N/A	N/A
9125988	12/2014	Karlsson	N/A	N/A
9132235	12/2014	Holmqvist	N/A	N/A
9199038	12/2014	Daniel	N/A	N/A
9205199	12/2014	Kemp et al.	N/A	N/A
9216256	12/2014	Olson et al.	N/A	N/A
9233213	12/2015	Olson et al.	N/A	N/A
9233214	12/2015	Kemp et al.	N/A	N/A
9233215	12/2015	Hourmand et al.	N/A	N/A
9242044	12/2015	Markussen	N/A	N/A
9272098	12/2015	Hourmand et al.	N/A	N/A
9283326	12/2015	Kemp et al.	N/A	N/A
9283327	12/2015	Hourmand et al.	N/A	N/A
9283328	12/2015	Dasbach	N/A	N/A
9308327	12/2015	Marshall et al.	N/A	N/A
9333304	12/2015	Brereton et al.	N/A	N/A
9339607	12/2015	Langley et al.	N/A	N/A
9352088	12/2015	Ekman et al.	N/A	N/A
9358345	12/2015	Brereton et al.	N/A	N/A
9358351	12/2015	Ekman et al.	N/A	N/A
9393368	12/2015	Nzike et al.	N/A	N/A
9402957	12/2015	Adams et al.	N/A	N/A
9408976	12/2015	Olson et al.	N/A	N/A
9408977	12/2015	Butler et al.	N/A	N/A
9408979	12/2015	Veasey et al.	N/A	N/A
9415165	12/2015	Cowe	N/A	N/A
9421336	12/2015	Ekman et al.	N/A	N/A
9427525	12/2015	Barrow-Williams et al.	N/A	N/A
9446196	12/2015	Hourmand et al.	N/A	N/A
9446201	12/2015	Holmqvist	N/A	N/A
9457149	12/2015	Kemp et al.	N/A	N/A
9457152	12/2015	Raab et al.	N/A	N/A
9492622	12/2015	Brereton et al.	N/A	N/A
9662452	12/2016	Daniel	N/A	N/A
9867940	12/2017	Holmqvist et al.	N/A	N/A
9872961	12/2017	Fourt et al.	N/A	N/A
10118001	12/2017	Fourt et al.	N/A	N/A
10314981	12/2018	Sampson et al.	N/A	N/A
10350362	12/2018	Dennis, Jr. et al.	N/A	N/A
10363377	12/2018	Atterbury et al.	N/A	N/A
10420898	12/2018	Daniel	N/A	N/A
10569019	12/2019	Hirschel et al.	N/A	N/A
RE47903	12/2019	Hourmand et al.	N/A	N/A
10799647	12/2019	Hosteettler et al.	N/A	N/A

RE48593	12/2020	Hourmand et al.	N/A	N/A
11298462	12/2021	Atterburg et al.	N/A	N/A
11383044	12/2021	Tschirren et al.	N/A	N/A
11400217	12/2021	Hourmand et al.	N/A	N/A
11452821	12/2021	Lafever et al.	N/A	N/A
11458252	12/2021	Hourmand et al.	N/A	N/A
11471601	12/2021	Hourmand et al.	N/A	N/A
11607495	12/2022	Hourmand et al.	N/A	N/A
11612691	12/2022	Hourmand et al.	N/A	N/A
12329936	12/2024	Hourmand et al.	N/A	N/A
12337146	12/2024	Hourmand et al.	N/A	N/A
12337147	12/2024	Hourmand et al.	N/A	N/A
12337148	12/2024	Hourmand et al.	N/A	N/A
2002/0007154	12/2001	Hansen et al.	N/A	N/A
2002/0095120	12/2001	Larsen et al.	N/A	N/A
2003/0105430	12/2002	Lavi et al.	N/A	N/A
2004/0039336	12/2003	Amark et al.	N/A	N/A
2005/0101919	12/2004	Brunnberg	N/A	N/A
2005/0203466	12/2004	Hommann et al.	N/A	N/A
2005/0222539	12/2004	Gonzales et al.	N/A	N/A
2006/0153693	12/2005	Fiechter et al.	N/A	N/A
2006/0224124	12/2005	Scherer	N/A	N/A
2006/0258990	12/2005	Weber	N/A	N/A
2006/0270984	12/2005	Hommann	N/A	N/A
2006/0270985	12/2005	Hommann et al.	N/A	N/A
2006/0287630	12/2005	Hommann	N/A	N/A
2007/0027430	12/2006	Hommann	N/A	N/A
2007/0129686	12/2006	Daily et al.	N/A	N/A
2008/0262427	12/2007	Hommann	N/A	N/A
2008/0269692	12/2007	James et al.	N/A	N/A
2008/0312591	12/2007	Harrison	N/A	N/A
2009/0012471	12/2008	Harrison	N/A	N/A
2009/0088688	12/2008	Timothy Donald et al.	N/A	N/A
2009/0292246	12/2008	Slate et al.	N/A	N/A
2009/0312705	12/2008	Grunhut et al.	N/A	N/A
2010/0049125	12/2009	James et al.	N/A	N/A
2010/0185178	12/2009	Sharp et al.	N/A	N/A
2010/0256570	12/2009	Maritan	N/A	N/A
2010/0262083	12/2009	Grunhut et al.	N/A	N/A
2010/0268170	12/2009	Carrel	604/198	A61M 5/2033
2010/0280461	12/2009	Forstreuter	29/428	A61M 5/31515
2010/0312195	12/2009	Johansen et al.	N/A	N/A
2011/0046567	12/2010	Radmer	604/218	A61M 5/31515
2011/0270161	12/2010	Harrison et al.	N/A	N/A
2012/0010575	12/2011	Jones et al.	N/A	N/A

2012/0041387	12/2011	Bruggemann et al.	N/A	N/A
2012/0053528	12/2011	Bollenbach et al.	N/A	N/A
2012/0116319	12/2011	Grunhut	N/A	N/A
2012/0172817	12/2011	Bruggemann et al.	N/A	N/A
2013/0035647	12/2012	Veasey et al.	N/A	N/A
2013/0041328	12/2012	Daniel	N/A	N/A
2013/0123710	12/2012	Ekman et al.	N/A	N/A
2013/0261556	12/2012	Jones et al.	N/A	N/A
2013/0274662	12/2012	Hourmand et al.	N/A	N/A
2013/0274666	12/2012	Brereton et al.	N/A	N/A
2013/0274677	12/2012	Ekman et al.	N/A	N/A
2013/0289525	12/2012	Kemp et al.	N/A	N/A
2013/0310757	12/2012	Brereton et al.	N/A	N/A
2013/0345643	12/2012	Hourmand et al.	N/A	N/A
2015/0273157	12/2014	Kohlbrenner et al.	N/A	N/A
2016/0089498	12/2015	Daniel	N/A	N/A
2018/0064875	12/2017	Holmqvist	N/A	N/A
2019/0374717	12/2018	Swanson et al.	N/A	N/A
2022/0211947	12/2021	Hourmand et al.	N/A	N/A
2022/0218906	12/2021	Hourmand et al.	N/A	N/A
2022/0313915	12/2021	Hourmand et al.	N/A	N/A
2023/0019806	12/2022	Hourmand et al.	N/A	N/A
2023/0086760	12/2022	Hourmand et al.	N/A	N/A
2024/0207520	12/2023	Hourmand et al.	N/A	N/A
2024/0238521	12/2023	Hourmand et al.	N/A	N/A
2024/0238522	12/2023	Hourmand et al.	N/A	N/A
2024/0238523	12/2023	Hourmand et al.	N/A	N/A
2025/0152825	12/2024	Hourmand et al.	N/A	N/A
2025/0152826	12/2024	Hourmand et al.	N/A	N/A
2025/0152828	12/2024	Hourmand et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
19819409	12/1998	DE	N/A
202004016787	12/2004	DE	N/A
102004060146	12/2004	DE	N/A
202007000578	12/2006	DE	N/A
102005052502	12/2006	DE	N/A
0666084	12/1994	EP	N/A
0824923	12/1997	EP	N/A
0991441	12/2002	EP	N/A
2399634	12/2010	EP	N/A
2468334	12/2011	EP	N/A
2468335	12/2011	EP	N/A
2438592	12/2006	GB	N/A
2001-521792	12/2000	JP	N/A
2002-528182	12/2001	JP	N/A

2008-521482	12/2007	JP	N/A
2010-520786	12/2009	JP	N/A
2010-540059	12/2009	JP	N/A
WO 1999/022790	12/1998	WO	N/A
WO 1999/022792	12/1998	WO	N/A
WO 1999/053979	12/1998	WO	N/A
WO 2000/024441	12/1999	WO	N/A
WO 2002/047746	12/2001	WO	N/A
WO 2003/062672	12/2002	WO	N/A
WO 2005/044345	12/2004	WO	N/A
WO 2005/097238	12/2004	WO	N/A
WO 2005/115507	12/2004	WO	N/A
WO 2006/057604	12/2005	WO	N/A
WO 2006/111861	12/2005	WO	N/A
WO 2007/083115	12/2006	WO	N/A
WO 2007/099044	12/2006	WO	N/A
WO 2007/129324	12/2006	WO	N/A
WO 2008/059385	12/2007	WO	N/A
WO 2008/112472	12/2007	WO	N/A
WO 2008/116688	12/2007	WO	N/A
WO 2009/040602	12/2008	WO	N/A
WO 2009/040607	12/2008	WO	N/A
WO 2009/040672	12/2008	WO	N/A
WO 2009/062508	12/2008	WO	N/A
WO 2010/035060	12/2009	WO	N/A
WO 2010/063707	12/2009	WO	N/A
WO 2010/136077	12/2009	WO	N/A
WO 2011/012903	12/2010	WO	N/A
WO 2011/109205	12/2010	WO	N/A
WO 2011/111006	12/2010	WO	N/A
WO 2011/117592	12/2010	WO	N/A
WO 2011/126439	12/2010	WO	N/A
WO 2012/045350	12/2011	WO	N/A
WO 2012/085021	12/2011	WO	N/A
WO 2012/085024	12/2011	WO	N/A
WO 2019/074788	12/2018	WO	N/A
WO 2020/190529	12/2019	WO	N/A
WO 2021/008839	12/2020	WO	N/A

OTHER PUBLICATIONS

International Preliminary Report on Patentability in International Appln. No. PCT/EP2011/073502, dated Jun. 25, 2013, 7 pages. cited by applicant

International Search Report and Written Opinion in International Appln. No. PCT/EP2011/073502, dated Mar. 19, 2012, 9 pages. cited by applicant

Primary Examiner: Carpenter; William R

Attorney, Agent or Firm: Fish & Richardson P.C.

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 18/498,416, filed Oct. 31, 2023, which is a continuation of U.S. patent application Ser. No. 16/909,107, filed Jun. 23, 2020, now U.S. Pat. No. 11,833,331, which is a continuation of U.S. patent application Ser. No. 15/669,571, filed Aug. 4, 2017, now U.S. Pat. No. 10,729,853, which is a continuation of U.S. patent application Ser. No. 13/993,528, with a 371 filing date of Jun. 12, 2013, now U.S. Pat. No. 9,724,472, which is a U.S. National Phase Applications pursuant to 35 U.S.C. § 371 of International Application No. PCT/EP2011/073502, filed Dec. 21, 2011, which claims priority to European Patent Application No. 10196067.2, filed Dec. 21, 2010, and U.S. Provisional Patent Application No. 61/432,255, filed Jan. 13, 2011. The entire disclosure contents of these applications are herewith incorporated by reference into the present application.

TECHNICAL FIELD

(1) The invention relates to an auto-injector for administering a dose of a liquid medicament.

BACKGROUND

(2) Administering an injection is a process which presents a number of risks and challenges for users and healthcare professionals, both mental and physical.

(3) Injection devices (i.e. devices capable of delivering medicaments from a medication container) typically fall into two categories—manual devices and auto-injectors.

(4) In a manual device—the user must provide the mechanical energy to drive the fluid through the needle. This is typically done by some form of button/plunger that has to be continuously pressed by the user during the injection. There are numerous disadvantages to the user from this approach. If the user stops pressing the button/plunger then the injection will also stop. This means that the user can deliver an underdose if the device is not used properly (i.e. the plunger is not fully pressed to its end position). Injection forces may be too high for the user, in particular if the patient is elderly or has dexterity problems.

(5) The extension of the button/plunger may be too great. Thus it can be inconvenient for the user to reach a fully extended button. The combination of injection force and button extension can cause trembling/shaking of the hand which in turn increases discomfort as the inserted needle moves.

(6) Auto-injector devices aim to make self-administration of injected therapies easier for patients. Current therapies delivered by means of self-administered injections include drugs for diabetes (both insulin and newer GLP-1 class drugs), migraine, hormone therapies, anticoagulants etc.

(7) Auto-injectors are devices which completely or partially replace activities involved in parenteral drug delivery from standard syringes. These activities may include removal of a protective syringe cap, insertion of a needle into a patient's skin, injection of the medicament, removal of the needle, shielding of the needle and preventing reuse of the device. This overcomes many of the disadvantages of manual devices. Injection forces/button extension, hand-shaking and the likelihood of delivering an incomplete dose are reduced. Triggering may be performed by numerous means, for example a trigger button or the action of the needle reaching its injection depth. In some devices the energy to deliver the fluid is provided by a spring.

(8) US 2002/0095120 A1 discloses an automatic injection device which automatically injects a pre-measured quantity of fluid medicine when a tension spring is released. The tension spring moves an ampoule and the injection needle from a storage position to a deployed position when it is released. The content of the ampoule is thereafter expelled by the tension spring forcing a piston forward inside the ampoule. After the fluid medicine has been injected, torsion stored in the tension spring is released and the injection needle is automatically retracted back to its original storage position.

SUMMARY

(9) It is an object of the present invention to provide an improved auto-injector.

(10) The object is achieved by an auto-injector according to claim 1.

(11) Preferred embodiments of the invention are given in the dependent claims.

(12) In the context of this specification, the terms distal and proximal are defined from the point of view of a person receiving an injection. Consequently, a proximal direction refers to a direction pointing towards the body of a patient receiving the injection and a proximal end defines an end of an element that is directed towards the body of the patient. Respectively, the distal end of an element or the distal direction is directed away from the body of the patient receiving the injection and opposite to the proximal end or proximal direction.

(13) According to the invention, an auto-injector for administering a dose of a liquid medicament comprises a substantially cylindrical housing arranged to contain a pre-filled syringe filled with the medicament, a needle shroud slidably arranged with respect to the housing and adapted to rest on the skin of a patient receiving an injection, a releasable drive means arranged within the housing that is capable of, upon release, translating the needle shroud in a proximal direction towards a safe position and a rotating collar rotatably arranged within the housing. The needle shroud in the safe position surrounds the injection needle after the injection has been carried out. The rotating collar engages the needle shroud in a manner that forces the rotating collar to rotate within the housing when the needle shroud is translated in the proximal direction.

(14) A stopper may be arranged in the syringe barrel arranged to seal its distal end and to expel the dose of medicament. A coupling shroud may be slidably arranged within the housing and releasably coupled to a plunger that is connected to the stopper. The drive means may be arranged between a distal end of the housing and the coupling shroud to bias the coupling shroud in a proximal direction towards the skin of a patient receiving an injection.

(15) The rotating collar creates friction to slow down a proximal movement of the needle shroud that rests on the skin of the patient during the injection. The rotating collar acts as a dampening element that alleviates the pressure exerted upon the skin of the patient by the needle shroud that is driven by the drive means. Thus, the risk of injuries is reduced and, in particular, bruises may be avoided. Furthermore, the modulus of resilience of the single drive means may be chosen to be sufficiently large without having to worry about potential injury risks. Thus, the modulus of resilience of the drive means is adapted to reliably provide an energy supply for executing a plurality of actions comprising, among others, the advancing and releasing of the needle shroud, the displacement of the stopper to expel the medicament and the decoupling of the plunger and the coupling shroud.

(16) Preferably, the rotating collar comprises a pin that engages a helical recess formed into the needle shroud. The engagement of the helical recess and the pin forces the rotating collar to rotate around the needle shroud when the needle shroud is translated. This dampens the proximal movement of the needle shroud and thus reduces the pressure exerted upon the skin of the patient by generating friction.

(17) According to a possible embodiment of the invention, the needle shroud is slidable in a distal direction from an advanced position to a retracted position. In particular, the needle shroud may be slid to the retracted position by placing the auto-injector onto the skin of the patient receiving the injection and pressing the needle shroud against the skin of the patient. The needle shroud in the retracted position indicates the correct placement of the auto-injector. An activation of a mechanism of the auto-injector delivering the medicament to the patient requires the needle shroud to be positioned in the retracted position to ensure that the auto-injector is properly used. The needle shroud is slidable in the proximal direction from the retracted position to the safe position, wherein the needle shroud surrounds an injection needle of the pre-filled syringe to prevent accidental needle stick injuries after the injection has been carried out.

(18) According to another possible embodiment of the invention, the drive means is arranged to be released by manual actuation of a release element that is hinged to a lateral side of the housing.

Conventional auto-injectors are commonly activated by actuating a push button or the like arranged at a distal end of the auto-injector. An inexperienced user of such a conventional auto-injector may easily mistake the distal end for a proximal end of the auto-injector and thus may pierce his finger while trying to actuate the auto-injector. The lateral arrangement of the release element is a simple means to prevent such accidental needle stick injuries resulting from a misuse of the auto-injector.

(19) Alternatively, the release element is in slidable arrangement with the housing and may be translated with respect to the housing in the proximal direction to release the drive means. The auto-injector according to this embodiment of the invention is particularly intuitive to operate.

(20) According to another possible embodiment of the invention, the auto-injector comprises safety means that cooperate with the needle shroud that is arranged to prevent a release of the drive means when the needle shroud is in the advanced position and hence is not pushed against the skin of the patient. This mechanism avoids an early release of the drive means and thus a premature expelling of the medicament. Furthermore, injuries resulting from an activation of the drive means when the auto-injector is not placed, or not properly placed onto the skin of the patient are reduced.

(21) The safety means may comprise a blocking element slidably arranged relative to the housing. The blocking element is arranged to limit a pivoting movement of the release element hinged to the lateral side of the housing when the needle shroud is in the advanced position. A release of the drive means is thus prevented. The release element is allowed to pivot about the hinge when the needle shroud is moved to the retracted position by pressing the needle shroud against the skin surface of the patient receiving the injection.

(22) In one possible embodiment of the invention, the safety means comprises an elastic bushing that engages a plunger of the pre-filled syringe and/or a coupling shroud that is biased by the drive means and coupled to the plunger of the pre-filled syringe. The elastic bushing is firmly attached to a proximal end of the housing and may engage the coupling shroud and/or the plunger to prevent an inadvertent release of the drive means.

(23) In another preferred embodiment of the invention, the drive means is capable of, upon release, driving the coupling shroud releasably coupled to the plunger from a first position in the proximal direction. The coupling shroud is driven by the drive means to interact with different components of the auto-injector, so that the resilient force provided by the drive means may be used to power a variety of functions of the auto-injector. The proximal translatory movement of the coupling shroud with respect to the housing translates a syringe retainer receiving the pre-filled syringe in the proximal direction to expose the injection needle of the pre-filled syringe, depresses the plunger connected to a stopper into the syringe barrel to expel the dose of medicament and translates the needle shroud in the proximal direction. The coupling shroud mediates the resilient force provided by the drive means to this variety of different components of the auto-injector and thus allows for a compact design of the auto-injector. In particular, the auto-injector may be powered by just a single drive means to accomplish a plurality of tasks necessary for safely carrying out the injection.

(24) According to yet another possible embodiment of the invention, the syringe retainer is releasably mounted to the housing. The needle shroud in the retracted position releases the syringe retainer to allow for the proximal translation of the syringe retainer with respect to the housing. Thus, a proximal movement of the syringe retainer that in particular inserts the injection needle into the skin of the patient is prevented until the auto-injector is correctly placed upon the skin of the patient and the needle shroud is pushed against the skin surface towards the retracted position. This prevents an inadvertent early release of the drive means.

(25) The coupling shroud is initially in a first position, and coupled to the plunger to translate the syringe retainer proximally, whereby the injection needle is inserted into the skin of the patient, and to depress the stopper into the syringe barrel to expel the medicament. The plunger and the coupling shroud are decoupled from each other at the proper pre-determined second position after the medicament has been completely or partially delivered. The pre-determined second position is defined by a longitudinal aperture in the housing.

(26) According to another possible embodiment of the invention, a coupling catch is arranged to abut against a shoulder formed to the plunger as a particularly simple and reliable means to releasably couple the plunger to the coupling shroud. The coupling shroud is moved by the action of the relaxing drive means in the proximal direction and is coupled to the plunger connected to the stopper to insert the injection needle before the injection and to expel the medication during the injection.

(27) The aperture formed into the lateral side of the housing at the second position allows the coupling catch to deflect radially outwards at the second position, so that the coupling shroud is decoupled from the plunger after the medicament is partially or completely delivered.

(28) Preferably, the drive means is arranged as a single compression spring. The mechanism of the auto-injector is arranged in a manner that a plurality of functions is executed by the single drive means. The injection needle is inserted into the skin of the patient, the plunger is translated to expel the medicament and the needle shroud is moved proximally to provide needle safety after the injection is completed by the action of the spring means. Conventional auto-injectors usually comprise a plurality of spring means to accomplish these tasks. The auto-injector according to the invention comprises only few parts and is particularly inexpensive to mass-produce. Consequently, the auto-injector is particularly suited as a single-use device that may be disposed after an injection has been carried out.

(29) The auto-injector may preferably be used for subcutaneous or intra-muscular injection, particularly for delivering one of an analgetic, an anticoagulant, insulin, an insulin derivate, heparin, Lovenox, a vaccine, a growth hormone, a peptide hormone, a proteine, antibodies and complex carbohydrates.

(30) Further scope of applicability of the present invention will become apparent from the detailed description given hereinafter. However, it should be understood that the detailed description and specific examples, while indicating preferred embodiments of the invention, are given by way of illustration only, since various changes and modifications within the spirit and scope of the invention will become apparent to those skilled in the art from this detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present invention will become more fully understood from the detailed description given hereinbelow and the accompanying drawings which are given by way of illustration only, and thus, are not limitive of the present invention, and wherein:

(2) FIGS. 1A und 1B show two different sectional views of an auto-injector according to a first embodiment of the invention with a hinged release element before an injection,

(3) FIG. 2 is an expanded sectional view of the auto-injector according to the first embodiment, wherein the release element is blocked to prevent an inadvertent release of a drive means,

(4) FIG. 3 is an expanded sectional view of the hinged release element that is actuated to release the drive means,

(5) FIGS. 4A and 4B are two different sectional views of the auto-injector according to the first embodiment after a drug has been delivered,

(6) FIG. 5 is an isometric view of a needle shroud,

(7) FIGS. 6A and 6B are two different sectional views of the auto-injector according to the first embodiment after an injection has been performed,

(8) FIGS. 7A and 7B are two different sectional views of an auto-injector according to a second embodiment of the invention,

(9) FIG. 8 is a sectional view of a distal end section of the auto-injector according to the second embodiment of the invention,

(10) FIGS. **9A** and **9B** are two different sectional views of an auto-injector according to a third embodiment of the invention,

(11) FIG. **10** is a sectional view of a distal end section of the auto-injector according to the third embodiment of the invention.

(12) Corresponding parts are marked with the same reference symbols in all figures.

DETAILED DESCRIPTION

(13) FIGS. **1A** and **1B** show two sectional views of an essentially cylindrical auto injector **1** according to a first embodiment of the invention, wherein the sectional plane shown in FIG. **1A** is oriented perpendicularly to the one shown in FIG. **1B**. The auto-injector **1** comprises a housing **2**, a proximal needle shroud or sleeve **3**, a syringe retainer **4** adapted to mount and move translatably with a pre-filled syringe **5** within the housing **2**, a coupling/decoupling shroud or member **6** slidably arranged within the housing **2** and a release element **7** hinged to a lateral side of the substantially cylindrical housing **2** of the auto-injector **1**.

(14) A single drive means **8** is arranged between the distal end of the housing **2** and the coupling shroud **6** to bias the coupling shroud **6** in a proximal direction P towards the skin of a patient receiving an injection.

(15) According to one possible embodiment of the invention, the drive means **8** is arranged as a single, conventional compression spring.

(16) The coupling shroud **6** is releasably coupled to a plunger **9** that is connected to a stopper **10** fluid-tightly sealing a distal end of a syringe barrel **11** containing a dose of a medicament M. An inner cavity of the syringe barrel **11** is in fluid communication with an injection needle **12**, so that the dose of the medicament M may be expelled through the injection needle **12** by displacing the stopper **10** in the proximal direction P.

(17) Before the injection, the coupling shroud **6** abuts against a distal end of the release element **7** to releasably retain the coupling shroud **6** in a first position I, wherein the coupling shroud **7** is located at a distal end of the housing **2**. The drive means **8** is compressed, so that the coupling shroud **6** is strongly biased in the proximal direction P.

(18) The plunger **9** extends from the syringe barrel **11** in a distal direction D and comprises a shoulder **9.1** (e.g., an enlarged shoulder) of increased diameter. The coupling shroud **6** comprises an inwardly protruding coupling catch **6.1** that bears against the shoulder **9.1** so that the plunger **9** and the coupling shroud **6** may be jointly moved in the proximal direction P by the action of the relaxing drive means **8**.

(19) The proximal end of the needle shroud **3** is designed to be pushed against the skin surface of the patient during the injection. Edges of the needle shroud **3** may thus be smoothed to avoid injuries. The needle shroud **3** is slidably arranged within the housing **2** of the auto-injector **1**, so that the needle shroud **3** may be pushed from an advanced position PA shown in FIGS. **1A** and **1B** in the distal direction D. A biasing means **13** bears against the needle shroud **3** and the housing **2** to bias the needle shroud **3** towards the advanced position PA.

(20) An annular rotating collar **14** engages an outer surface of the needle shroud **3**. The rotating collar **14** rotates around an axis of the substantially cylindrical auto-injector **1** when the needle shroud **3** is longitudinally displaced in the proximal and/or the distal direction P, D. The rotating collar **14** acts as a damping means that creates friction to slow down the movement of the needle shroud **3** and to reduce the pressure exerted onto the skin of the patient receiving the injection.

(21) The release element **7** hinged to the housing **2** works like a see-saw: a proximal section may be pushed radially inwards, whereby the release element **7** pivots about a hinge **15**, so that the distal section of the release element **7** moves radially outwards and the coupling shroud **6** is disengaged to release the drive means **8**.

(22) The auto-injector **1** comprises safety means S that prevent an early release of the drive means **8**. The safety means S ensure that the needle shroud **3** is pushed against the skin of the person receiving the injection before the drive means **8** may be released.

(23) According to the first embodiment of the invention, the safety means S comprise a blocking element **16** slidably arranged with the housing **2**. When the needle shroud **3** is positioned in the advanced position PA, the blocking element **16** is positioned to prevent a pivoting movement of the release element **7** and thus a release of the coupling shroud **6**. A radially outwards protruding blocking projection **16.1** of the blocking element **16** is located opposite to an inward protrusion **7.1** formed to the release element **7**. If the proximal section of the release element **7** is pushed inwards, the inward protrusion **7.1** abuts against the blocking projection **16.1** to limit the pivoting movement of the release element **7**, so that a release of the coupling shroud **6** and the drive means **8** is prevented.

(24) FIG. 2 shows a proximal section of the auto-injector **1** in a sectional view with the blocking element **16** positioned to prevent an inadvertent actuation of the hinged release element **7** to release the drive means **8**.

(25) A distal end of the needle shroud **3** is clipped to the housing **2** and retained between two inwardly protruding retaining protrusions or deflectable arms **2.1** formed into an inner surface of the housing **2**. The two retaining protrusions **2.1** are longitudinally displaced from each other to limit the range of axial displacement of the needle shroud **3** with respect to the housing **2**. A boss **3.1** formed into an outer surface of the needle shroud **3** bears against an inner surface of the blocking element **16**, so that the blocking element **16** may move with the needle shroud **3** in the proximal direction P to deblock the release element **7**.

(26) The blocking projection **16.1** comprises a central indentation that forces a user of the auto-injector **1** to perform a sequence of actions necessary to inject the dose of the medicament M in the proper order. If the release element **7** is pushed inwards before the needle shroud **3** is moved proximally from the advanced position PA towards a retracted position PR (see FIG. 3) by pushing the needle shroud **3** towards the skin of the patient, the inward projection **7.1** is retained in the central indentation of the blocking projection **16.1**, so that both the longitudinal displacement of the needle shroud **3** and the pivoting movement of the release element **7** is blocked.

(27) A proper sequence of actions for injecting the dose of the medicament M is described in the following. First, the user pushes the needle shroud **3** against the skin to move the needle shroud **3** distally to the retracted position PR illustrated in FIG. 3. The blocking element **16** jointly moves with the needle shroud **3** in the distal direction D, so that the release element **7** may be manually actuated to pivot about the hinge **15**, whereby the drive means **8** are released.

(28) Upon release of the drive means **8**, the coupling shroud **6** is urged in the proximal direction P. The single and fully compressed drive means **8** drives the coupling shroud **6** and the plunger **9** coupled thereto in the proximal direction P. The coupling shroud **6** first pushes the syringe retainer **4** by means of plunger **9**, stopper **10** and the friction between stopper **10** and syringe **11** proximally to insert the injection needle **12** into the skin of the patient and a first clip connection **2.2** formed into a lateral side of the housing **2** latches to an outward protrusion **4.1** of the syringe retainer **4**, as illustrated in more detail in FIG. 4B.

(29) The syringe retainer **4** and the pre-filled syringe **5** mounted thereto is now locked to the housing **2**. The coupling shroud **6** is moved further in the proximal direction P by the action of the relaxing drive means **8**, whereby the plunger **9** is depressed into the syringe barrel **11** to expel the dose of the medicament M contained therein through the injection needle **12**.

(30) FIGS. 4A and 4B show two sectional views of the auto-injector **1** according to the first embodiment of the invention with the plunger **9** fully depressed within the syringe barrel **11**. The dose of the medicament M has been delivered beneath the skin of the patient. The coupling shroud **6** is located in an intermediate second position II. The drive means **8** is not yet completely discharged and biases the coupling shroud **6** in the proximal direction P. The shoulder **9.1** engages a ramp of the coupling catch **6.1** to deflect the coupling catch **6.1** in a radial outward direction. An aperture **2.3** is formed into the housing **2** to allow for a radial outward deflection of the coupling catch **6.1**, so that the coupling catch **6.1** overcomes the shoulder **9.1** decoupling the coupling

shroud **6** from the plunger **9**.

(31) In a possible embodiment of the invention, the aperture **2.3** defining the second position II is located at a longitudinal position along the housing **2** that allows for a full depression of the plunger **9** completely emptying the syringe barrel **11** before the plunger **9** is decoupled from the coupling shroud **6**.

(32) Alternatively, the aperture **2.3** defining the second position II may be located at a longitudinal position along the housing **2** that allows for an adjustment space accounting for manufacturing tolerances. The adjustment space is dimensioned as to allow for a reliable decoupling of the plunger **9** from the coupling shroud **6** even if the parts constituting the auto-injector **1** comprise mismatch in mould or are slightly misaligned. In this alternative embodiment, the dose of the medicament **M** may or may not be completely expelled before the plunger **9** is decoupled from the coupling shroud **6**.

(33) The retaining protrusions **2.1** are elastically supported and may be deflected radially outwards to release the needle shroud **3**. The coupling shroud **6** engages a ramp of the retaining protrusions **2.1** and splays the retaining protrusions **2.1** outwards, whereby the needle shroud **3** is released and allowed to move proximally from the retracted position PR towards an extended safe position PS.

(34) The drive means **8** is still partially loaded when the coupling shroud **6** is located in the second position II. In a possible embodiment of the invention the biasing force of the drive means **8** exerted on the coupling shroud **6** in the second position II is about 10 N.

(35) The coupling shroud **6** bears against a distal end of the needle shroud **3**, so that the needle shroud **3** may be moved to the safe position PS by the action of the further relaxing drive means **8**. As the biasing force exerted onto the needle shroud **3** by the drive means **8** may be relatively large and could even bruise the patient, the rotating collar **14** is arranged within the housing **2** to partially absorb the excess energy of the drive means **8** and slow down the proximal movement of the needle shroud **3** by generating friction.

(36) FIG. 5 shows an isometric view of the needle shroud **3**. A helical recess **3.2** is formed into a tubular proximal section **3.3** of the needle shroud **3**. The proximal section **3.3** of the needle shroud **3** is inserted into the annular rotating collar **14**, wherein a pin **14.1** formed to an inner surface of the rotating collar **14** protrudes into the helical recess **3.2** as shown in FIG. 6A. The linear translatory movement of needle shroud **3** towards the safe position PS thus causes the rotating collar **14** to rotate within the housing **2** around the axis of the auto-injector **1**.

(37) FIGS. 6A and 6B show two different sectional views of the auto-injector **1** according to the first embodiment of the invention after the injection has been performed. The needle shroud **3** is permanently locked to the safe position PS by a second clip connection **2.4** formed into the housing **2**. The needle shroud **3** surrounds the injection needle **12** and extends a suitable distance proximally beyond the needle tip to avoid accidental needle stick injuries after the auto-injector **1** has been used.

(38) FIGS. 7A and 7B show two different sectional views of an auto-injector **1** according to a second embodiment of the invention before the injection. The sectional plane shown in FIG. 7A is oriented perpendicularly to the sectional plane shown in FIG. 7B.

(39) The needle shroud **3** of the auto-injector **1** according to the second embodiment substantially extends over the axial length of the housing **2**. Before the injection, the needle shroud **3** is mounted to the housing **2** by the retaining protrusions **2.1** that protrude into orifices formed into a lateral side of the needle shroud **3**. The orifices comprise a longitudinal length that allows the needle shroud **3** to be slid from the advanced position PA to the retracted position PR.

(40) A retaining catch **2.5** is formed to an inner surface of the housing **2** and protrudes through an opening formed into the needle shroud **3** to releasably mount the syringe retainer **4** retaining the pre-filled syringe **5**. The retaining catch **2.5** comprises a bevelled ramp and is deflectable in a radial outward direction. The retaining catch **2.5** latches to the outward protrusion **4.1** formed to the outer surface of the syringe retainer **4** when the needle shroud **3** is in the advanced position PA.

(41) The needle shroud 3 abuts against the bevelled ramp of the retaining catch 2.5 when the needle shroud 3 is moved from the advanced position PA in the distal direction D, whereby the retaining catch 2.5 is deflected in a radial outward direction and disengages the outward protrusion 4.1, so that the syringe retainer 4 may be moved in the proximal direction P.

(42) The release element 7, shown in more detail in FIG. 8, is arranged as a push button and mounted to a distal end of the housing 2. The release element 7 may be pushed in the proximal direction P to release the drive means 8 when the needle shroud 3 is in the retracted position PR, whereas the release element 7 and thus the release of the drive means 8 is blocked when the needle shroud 7 is in the advanced position PA.

(43) According to the second embodiment of the invention, the safety means S that prevent the early release of the drive means 8 comprises clips 2.6 that may deflect in a radial outward direction and a bushing 17 locking the plunger 9 before use of the auto-injector 1.

(44) Before the auto-injector 1 is used, the clips 2.6 formed to the housing 8 latch to the release element 7. (See FIG. 7B) The clips 2.6 block the movement of the release element 7 in the proximal direction P, so that a manual actuation of the release element 7 is prevented as long as the needle shroud 3 is in the advanced position PA. A distal movement of the release element 7 is blocked by a first detent 2.7 engaging an inner surface of the release element 7.

(45) The clip 2.6 comprises a ramp that the needle shroud 3 engages when pushed distally from the advanced position PA to the retracted position PR, whereby the clip 2.6 is deflected radially outwards to disengage the needle shroud 3. The release element 7 may be pushed in the proximal direction P when the needle shroud 3 is in the retracted position PR.

(46) The plunger 9 comprises a proximal portion 9.3 having two-axially extending portions 9.3 having different outer dimensions (e.g., diameters) measured in a direction perpendicular to the longitudinal axis of the plunger 9 and different axial dimensions (e.g., lengths) measured along the longitudinal axis of the plunger 9. The plunger 9 comprises a distal portion 9.4 comprising a distal end 9.2 of increased diameter that is retained in the bushing 17 firmly attached to a distal end of the housing 2. The bushing 17 comprises an inner surface corresponding to the distal end 9.2 of the plunger 9 that engages the distal end 9.2 in a locked position L to lock the plunger 9 and the coupling shroud 6 coupled thereto to the housing 2 before use of the auto-injector 1. The bushing 17 abuts radially against an annular inner collar 7.2 of the release element 7 in the locked position L shown in FIGS. 7A and 7B. A radial outward deflection of the bushing 17 releasing the plunger 9 is thus prevented.

(47) FIG. 8 shows a sectional view of a distal end section of the auto-injector 1 according to the second embodiment of the invention. The needle shroud 3 is located in the retracted position PR and the release element 7 is pushed in the proximal direction P, so that the bushing 17 disengages the annular inner collar 7.2 of the release element 7. The bushing 17 is positioned in an unlocked position U and may deflect outwardly to release the plunger 9.

(48) Furthermore, the bushing 17 acts as a counter bearing for the drive means 8 bearing against the bushing 17 in the distal direction D.

(49) FIGS. 9A and 9B show two different sectional views of an auto-injector 1 according to a third embodiment of the invention, wherein the release element 7 is arranged as an outer sleeve extending over a substantial length of the auto-injector 1.

(50) According to the third embodiment of the invention, the safety means S that prevent the early release of the drive means 8 comprise clips 2.6, second and third detents 2.8, 7.3, a locking catch 6.2 formed to the coupling shroud 6 and the bushing 17 that comprises an inner sleeve 17.1 receiving a lug 7.4, wherein the locking catch 6.2 latches to a collar 17.2 of the inner sleeve 17.1.

(51) The release element 7 of the third embodiment is gripped by a user to perform the injection. When the needle shroud 3 is in the advanced position PA, the proximal displacement of the release element 7 is blocked by the clips 2.6 in a similar manner as in the second embodiment described herein above.

(52) Additionally, the release element is releasably retained in position before the injection by the second and the third detents **2.8**, **7.3** respectively formed to an outer surface of the housing **2** and to an inner surface of the release element **7**, wherein the second and third detents **2.8**, **7.3** comprise correspondingly shaped ramps facing each other.

(53) The bushing **17** of the third embodiment comprises the inner sleeve **17.1** that receives the lug **7.4** formed to an inner surface of the release element **7**. A proximal end of the lug **7.4** snugly fits in the central aperture of the inner sleeve **17.1**, so that an inward deflection of the inner sleeve **17.1** is prevented.

(54) The inner sleeve **17.1** comprises a collar **17.2**. An inwardly protruding locking catch **6.2** of the coupling shroud **6** latches to the collar **17.2** before use of the auto-injector **1** to releasably retain the coupling shroud **6** in the first position I.

(55) FIG. **10** shows a sectional view of a distal end section of the auto-injector **1** according to the third embodiment of the invention. The release element **7** is actuated and moved in the proximal direction P. The proximal end of the lug **7.4** disengages the inner sleeve **17.1** of the bushing **17**, so that the inner sleeve **17.1** may bend radially inwards, whereby the locking catch **6.2** disengages the collar **17.2** and releases the coupling shroud **8** and the drive means **8**.

Claims

1. An auto-injector comprising: a housing defining an opening for a syringe containing a medicament; a sleeve disposed within the housing, the sleeve configured to axially move from a first position in which a needle of the syringe extends from a proximal end of the auto-injector to a safe position in which the needle of the syringe does not extend from the proximal end of the auto-injector, wherein the auto-injector is configured such that a clip locks the sleeve in the safe position once the sleeve has reached the safe position; a plunger rod disposed within the housing such that a longitudinal axis of the plunger rod is aligned with a longitudinal axis of the housing, the plunger rod having a proximal portion that is centered on the longitudinal axis of the plunger rod along an axial dimension from an enlarged shoulder of the plunger rod to a proximal end of the plunger rod, the proximal end of the plunger rod being configured to engage and displace a stopper of the syringe to dispense the medicament from the syringe, wherein the proximal portion of the plunger rod comprises two axially-extending portions having different outer dimensions measured in a direction perpendicular to the longitudinal axis of the plunger rod and different axial dimensions measured along the longitudinal axis of the plunger rod, a first axially-extending portion of the two axially-extending portions is proximal to a second axially-extending portion of the two axially-extending portions, and an outer dimension of the first axially-extending portion is greater than an outer dimension of the second axially-extending portion; an annular rotatable collar axially fixed within the housing and configured to rotate relative to the sleeve to increase an amount of time between medicament delivery and the sleeve arriving in the safe position; a spring disposed within the housing for biasing the enlarged shoulder of the plunger rod in a proximal direction within the auto-injector; and a decoupling member configured to allow a decoupling of the sleeve from the housing during a final portion of medicament delivery.
2. The auto-injector of claim 1, wherein the auto-injector is configured such that the decoupling member proximally moves with the plunger rod during medicament delivery.
3. The auto-injector of claim 2, wherein the decoupling member and the plunger rod are distinct components of the auto-injector.
4. The auto-injector of claim 1, wherein the auto-injector is configured to allow a proximal end of the decoupling member to deflect a deflectable arm to allow the decoupling of the sleeve from the housing.
5. The auto-injector of claim 1, wherein the outer dimension of the first axially-extending portion is smaller than an outer dimension of the enlarged shoulder in the direction perpendicular to the

longitudinal axis of the plunger rod.

6. The auto-injector of claim 1, wherein the plunger rod comprises a distal portion that extends distally from the enlarged shoulder, the distal portion configured to be releasably held in an initial position within the housing before medicament delivery.

7. An auto-injector comprising: a housing defining an opening for a syringe containing a medicament; a sleeve configured to move to a safe position in which a needle of the syringe is circumferentially covered, wherein the auto-injector is configured such that a clip locks the sleeve in the safe position once the sleeve has reached the safe position; a plunger rod disposed within the housing such that a longitudinal axis of the plunger rod is aligned with a longitudinal axis of the housing, the plunger rod having a proximal portion that is centered on the longitudinal axis of the plunger rod along an axial dimension between a shoulder of the plunger rod and a proximal end of the plunger rod, the proximal end of the plunger rod being configured to engage and displace a stopper of the syringe to dispense the medicament from the syringe, the plunger rod having a distal portion extending distally from the shoulder of the plunger rod, wherein the proximal portion of the plunger rod has two axially-extending portions having different outer dimensions measured in a direction perpendicular to the longitudinal axis of the plunger rod and different axial dimensions measured along the longitudinal axis of the plunger rod, a first axially-extending portion of the two axially-extending portions is proximal to a second axially-extending portion of the two axially-extending portions, and an outer dimension of the first axially-extending portion is greater than an outer dimension of the second axially-extending portion; an annular rotatable collar configured to rotate relative to the sleeve to delay an arrival of the sleeve in the safe position after medicament delivery is complete; a spring disposed within the housing for biasing the shoulder of the plunger rod in a proximal direction within the auto-injector; and a decoupling member configured to allow the sleeve to be decoupled from the housing during a final portion of medicament delivery.

8. The auto-injector of claim 7, wherein the auto-injector is configured such that the decoupling member proximally moves with the plunger rod during medicament delivery.

9. The auto-injector of claim 7, wherein the decoupling member and the plunger rod are distinct components of the auto-injector.

10. The auto-injector of claim 7, comprising the syringe disposed within the housing.

11. The auto-injector of claim 7, wherein the housing is cylindrical.

12. The auto-injector of claim 7, wherein the outer dimension of the first axially-extending portion is smaller than an outer dimension of the shoulder in the direction perpendicular to the longitudinal axis of the plunger rod.

13. The auto-injector of claim 7, wherein the auto-injector is configured to allow a proximal end of the decoupling member to deflect a deflectable arm to allow the sleeve to be decoupled from the housing.

14. An auto-injector comprising: a housing defining an opening for a syringe containing a medicament; a sleeve disposed within the housing, the sleeve configured to move to a safe position in which a needle of the syringe is completely disposed within the auto-injector; a plunger rod disposed within the housing such that a proximal portion of the plunger rod extending proximally from a shoulder of the plunger rod is centered on a longitudinal axis of the sleeve, and a proximal end of the plunger rod is configured to engage and displace a stopper of the syringe to dispense the medicament from the syringe, wherein the proximal portion of the plunger rod comprises two axially-extending portions having different outer dimensions measured in a direction perpendicular to the longitudinal axis of the sleeve and different axial dimensions measured along the longitudinal axis of the sleeve, a first axially-extending portion of the two axially-extending portions is proximal to a second axially-extending portion of the two axially-extending portions, and an outer dimension of the first axially-extending portion is greater than an outer dimension of the second axially-extending portion; an annular rotatable collar disposed within the housing and configured to rotate relative to the sleeve to increase an amount of time between medicament

delivery and the sleeve arriving in the safe position; a spring disposed within the housing and configured such that upon a release of the plunger rod a proximally-directed force of the spring proximally drives the proximal end of the plunger rod to displace the stopper of the syringe and dispense the medicament from the syringe; and a decoupling member configured to allow the sleeve to be released from a first axial position within the housing during a final portion of medicament delivery.

15. The auto-injector of claim 14, comprising a clip configured to permanently lock the sleeve in the safe position to limit reuse of the auto-injector after medicament delivery.

16. The auto-injector of claim 14, wherein the annular rotatable collar is axially fixed to the housing and comprises a pin configured to rotationally couple the annular rotatable collar to the sleeve.

17. The auto-injector of claim 16, wherein the sleeve comprises a recess configured to receive the pin of the annular rotatable collar to rotationally couple the annular rotatable collar to the sleeve.

18. The auto-injector of claim 14, wherein the plunger rod comprises a distal portion configured to releasably hold the plunger rod in an initial position prior to the release of the plunger rod, the distal portion extending distally from the shoulder of the plunger rod.

19. The auto-injector of claim 14, wherein the auto-injector is configured to allow the syringe to be in an initial axial position relative to the housing until the syringe is proximally driven by the proximal end of the plunger rod.

20. The auto-injector of claim 14, wherein the outer dimension of the first axially-extending portion is smaller than an outer dimension of the shoulder in the direction perpendicular to the longitudinal axis of the sleeve.

21. The auto-injector of claim 14, wherein the auto-injector is configured such that the decoupling member proximally moves with the plunger rod during medicament delivery.

22. The auto-injector of claim 14, wherein the auto-injector is configured to allow a proximal end of the decoupling member to deflect a deflectable arm to allow the sleeve to be released from the first axial position within the housing.

23. An auto-injector comprising: a housing for receiving a syringe containing a medicament; a sleeve disposed within the housing, the sleeve configured to move to a safe position relative to the housing to limit access to a needle of the syringe; a plunger rod having a proximal portion centered at a longitudinal axis of the sleeve, the proximal portion extending from a shoulder of the plunger rod to a proximal end of the plunger rod, and the proximal end of the plunger rod being configured to engage and displace a stopper of the syringe to dispense the medicament from the syringe, wherein the proximal portion of the plunger rod comprises two axially-extending portions, a first axially-extending portion of the two axially-extending portions is proximal to a second axially-extending portion of the two axially-extending portions, and an outer dimension of the first axially-extending portion is greater than an outer dimension of the second axially-extending portion; an annular rotatable collar axially fixed within the housing and configured to rotate relative to the sleeve to increase an amount of time between medicament delivery and the sleeve arriving in the safe position; a spring disposed within the housing and distal of the annular rotatable collar, the spring configured to provide a force to rotate the annular rotatable collar relative to the sleeve after medicament delivery; and a decoupling member configured to allow the sleeve to be decoupled from the housing, the decoupling member configured to move with the plunger rod during medicament delivery, wherein the auto-injector is configured to allow the syringe to be in an initial axial position relative to the housing until the syringe is proximally driven by the proximal end of the plunger rod.

24. The auto-injector of claim 23, wherein the plunger rod comprises a distal portion configured for releasably holding the plunger rod in a first axial position against a proximally-directed biasing force.

25. The auto-injector of claim 24, wherein the distal portion extends distally from the shoulder of

the plunger rod.

26. The auto-injector of claim 23, comprising a clip configured to permanently lock the sleeve in the safe position to prevent reuse of the auto-injector after medicament delivery.

27. The auto-injector of claim 23, wherein the decoupling member and the plunger rod are distinct components of the auto-injector.

28. The auto-injector of claim 23, wherein the outer dimension of the first axially-extending portion is smaller than an outer dimension of the shoulder in a direction perpendicular to the longitudinal axis of the sleeve.

29. The auto-injector of claim 28, wherein the outer dimension of the first axially-extending portion and the outer dimension of the second axially-extending portion are measured in the direction perpendicular to the longitudinal axis.

30. The auto-injector of claim 23, wherein the auto-injector is configured to allow a proximal end of the decoupling member to deflect a deflectable arm to allow the sleeve to be decoupled from the housing.
